

A restriction argument for the connected-domain construction question in $\mathcal{NCFB}_n(\Omega)$

Candidate solution packet for arXiv:1910.10538

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Status. Candidate full solution to Question 1 as literally stated, subject to human review. The construction retains strong irreducibility, but does not retain the additional spectral equalities in Theorem 3.10 of the source paper.

Abstract

Jiang, Ji, and Wu ask whether one can construct an operator in $\mathcal{NCFB}_n(\Omega)$ when the bounded planar domain Ω is connected rather than simply connected. The answer to this literal construction question is yes. Take the paper’s strongly irreducible example on a disk containing Ω and restrict its Cowen–Douglas base domain to Ω . The only point needing proof is that the spanning axiom in the Cowen–Douglas definition survives restriction to a nonempty open subset; this follows from the identity theorem for a local holomorphic frame. All additional $\mathcal{CFB}/\mathcal{NCFB}$ conditions are properties of the same operator and block decomposition, and hence are unchanged.

1 Source question

The source is C. Jiang, K. Ji, and J. Wu, *Similarity Invariants of Essentially normal Cowen–Douglas Operators and Chern Polynomials*, arXiv:1910.10538v1, Section 6, PDF page 27 [1]. The question is:

“Can we construct an operator in $\mathcal{NCFB}_n(\Omega)$ when Ω is connected?”

The source context assumes throughout that Ω is a bounded, open, connected subset of $\mathbb{C}$.

OF THE THEOREM.

□

6. SUMMARY AND FURTHER RESEARCH

In Section 3, it is critical to find an operator in $\mathcal{NCFB}_n(\Omega)$ (Theorem 3.10) to show that $\mathcal{NCFB}_n(\Omega)$ contains many nontrivial operators when Ω is simply connected. It is natural to ask the following questions:

1. Can we construct an operator in $\mathcal{NCFB}_n(\Omega)$ when Ω is connected?
2. Can we generalize Proposition 3.13 to the case Ω is (2-)connected?

In 1984, Misra [50] introduced homogeneous operators. An operator $T \in \mathcal{L}(\mathcal{H})$ is homogeneous

Figure 1: The two questions at the end of the source paper. This packet answers Question 1.

2 Definitions and proof intuition

For a connected open set $U \subset \mathbb{C}$, an operator A belongs to $\mathcal{B}_m(U)$ when, for every $w \in U$, the operator $A - wI$ is onto, $\dim \ker(A - wI) = m$, and

$$\overline{\text{span}}\{\ker(A - wI) : w \in U\} = \mathcal{H}.$$

The source defines $\mathcal{NCFB}_n(U)$ as a subclass of the essentially normal operators in $\mathcal{CFB}_n(U)$, with an upper-triangular block decomposition and specified idempotents whose Calkin images are self-adjoint [1].

Proof intuition. Cowen–Douglas membership is local except for the spanning axiom. If a vector is orthogonal to every eigenvector over a nonempty open subset, pair it with a local holomorphic frame. The resulting scalar holomorphic functions vanish on an open set, so the identity theorem propagates their vanishing throughout the larger connected domain. Thus the smaller family of eigenspaces is already total. Starting with the source paper’s disk example, we may therefore forget all eigenvalues outside Ω without changing the operator. The flag relations, Property (H), essential normality, and the idempotents do not depend on which base domain we use.

3 The restriction lemma

Lemma 1 (Restriction of the Cowen–Douglas base). *Let $U \subset \mathbb{C}$ be connected and open, let $V \subset U$ be nonempty and open, and let $A \in \mathcal{B}_m(U)$. If V is connected (as in the usual definition of a Cowen–Douglas domain), then $A \in \mathcal{B}_m(V)$.*

Proof. For $w \in V$, surjectivity of $A - wI$ and the equality $\dim \ker(A - wI) = m$ follow immediately from $A \in \mathcal{B}_m(U)$. It remains to prove totality.

Suppose $h \in \mathcal{H}$ is orthogonal to $\ker(A - wI)$ for every $w \in V$. Fix $w_0 \in V$ and $u \in U$. Since an open connected subset of $\mathbb{C}$ is path connected, choose a path in U from w_0 to u . Cover its compact image by a finite chain of open disks

$$W_0, W_1, \dots, W_N \Subset U$$

such that $W_0 \cap V$ is nonempty and open, $u \in W_N$, and every $W_{j-1} \cap W_j$ is nonempty and open. Shrinking the disks if necessary, the Cowen–Douglas bundle has a holomorphic frame $\gamma_{j,1}, \dots, \gamma_{j,m}$ on each W_j .

On W_0 , each scalar holomorphic function $w \mapsto \langle \gamma_{0,k}(w), h \rangle$ vanishes on $W_0 \cap V$. The identity theorem makes it vanish on all of W_0 . Consequently h is orthogonal to every fiber over $W_0 \cap W_1$. Applying the identity theorem to a frame on W_1 gives the same conclusion throughout W_1 . Induction along the chain shows that $h \perp \ker(A - uI)$. As $u \in U$ was arbitrary, h is orthogonal to all Cowen–Douglas fibers over U ; their totality gives $h = 0$. Hence the fibers over V are total. $\square$

4 Full answer to Question 1

Theorem 2. *Let $\Omega \subset \mathbb{C}$ be a bounded, nonempty, connected open set and let $n \geq 1$. Then $\mathcal{NCFB}_n(\Omega)$ is nonempty. In fact, it contains a strongly irreducible operator.*

Proof. The unit-disk part of Theorem 3.10 in the source paper constructs a strongly irreducible operator $A \in \mathcal{NCFB}_n(\mathbb{D})$ [1]. Choose $b \in \mathbb{C}$ and $R > 0$ such that $\Omega \subset U := b + R\mathbb{D}$, and set

$$T = bI + RA.$$

We first record directly that $T \in \mathcal{NCFB}_n(U)$. The Cowen–Douglas conditions transfer under $w = b + Rz$, because

$$T - wI = R(A - zI), \quad \ker(T - wI) = \ker(A - zI).$$

Essential normality is preserved since

$$[T^*, T] = R^2[A^*, A] \in \mathcal{K}(\mathcal{H}).$$

Use for T the same upper-triangular block decomposition and idempotents as for A . If $A_{k,j} = C_{k,j}A_{k,k+1} \cdots A_{j-1,j}$, then for $k < j$

$$T_{k,j} = RA_{k,j} = R^{1-(j-k)}C_{k,j}T_{k,k+1} \cdots T_{j-1,j},$$

and the relevant commutants are unchanged by the affine transformation. For adjacent diagonal blocks,

$$\tau_{T_{j,j}, T_{j+1,j+1}} = R \tau_{A_{j,j}, A_{j+1,j+1}},$$

so both the kernel and the range in Property (H) are unchanged. The Calkin conditions on the idempotents are literally the same. Thus all defining $\mathcal{CFB}_n$ and $\mathcal{NCFB}_n$ conditions persist. Finally, $\{T\}' = \{A\}'$, so T is strongly irreducible.

Apply Lemma 1 first to $T \in \mathcal{B}_n(U)$ and then to each diagonal block $T_{j,j} \in \mathcal{B}_1(U)$. It follows that $T \in \mathcal{B}_n(\Omega)$ and $T_{j,j} \in \mathcal{B}_1(\Omega)$. Every remaining $\mathcal{CFB}_n/\mathcal{NCFB}_n$ condition uses only T , its fixed block decomposition, Property (H), essential normality, and the same idempotents; none depends on enlarging or shrinking the Cowen–Douglas base domain. Therefore $T \in \mathcal{NCFB}_n(\Omega)$, and its strong irreducibility was already established. $\square$

5 Verification and limitations

- **Hypothesis check.** Boundedness of Ω is part of the source paper’s standing setup and ensures that a containing disk exists. For an unbounded Ω , no bounded operator can lie in $\mathcal{B}_n(\Omega)$, since every $w \in \Omega$ would be an eigenvalue.
- **Cowen–Douglas check.** Lemma 1 separately verifies the only nonlocal axiom, totality. The other two axioms are inherited pointwise.
- **Structure check.** The proof explicitly tracks affine scaling of the upper-triangular products and of the Rosenblum maps used in Property (H); the idempotents and their Calkin images do not change.
- **What is not claimed.** The constructed operator has the spectrum of the containing-disk model, not necessarily $\overline{\Omega}$. Hence this packet does not extend all three extra conclusions of source Theorem 3.10 to multiply connected domains. It answers the final section’s literal Question 1, and it does not address Question 2.

6 Bounded novelty check

On 27 August 2026, the run-wide registry, solution, attempt, and proof-gap indexes were searched for arXiv:1910.10538, the exact title, $\mathcal{NCFB}_n$, and connected-domain Cowen–Douglas keywords. A web/arXiv search used the exact question and the queries “NCFB Cowen Douglas connected”, the exact source title with “citations”, and “Cowen–Douglas NCFB connected domain”. This located the 2022 journal publication and later papers on Cowen–Douglas flag classes, but no later source explicitly answering Question 1 or presenting this restriction argument. The search is bounded and cannot certify novelty; the mathematical claim should be reviewed independently, especially because the argument is short.

7 Human-review recommendation

Check three points: (i) the analytic-continuation step in Lemma 1; (ii) the scaling of condition (2) in the $\mathcal{CFB}_n$ definition and of Property (H); and (iii) that the intended reading of Question 1 is its literal nonemptiness/construction statement, rather than the stronger spectrum-preserving analogue of Theorem 3.10. If those checks pass, Theorem 2 is a complete affirmative answer to Question 1.

References

- [1] C. Jiang, K. Ji, and J. Wu, *Similarity Invariants of Essentially normal Cowen–Douglas Operators and Chern Polynomials*, arXiv:1910.10538v1 (2019); Israel J. Math. **248** (2022), 229–270, <https://arxiv.org/abs/1910.10538>.